TOPIC 8

Ridge Regression: Explained

The original class slides, with a plain explanation under each one

> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

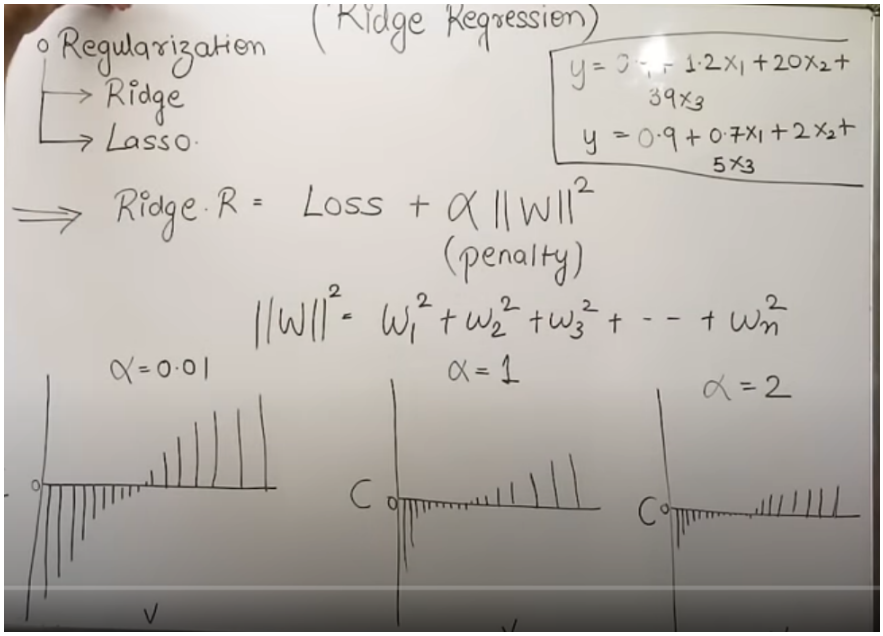
Regularization is a way to stop a model from becoming too complicated, by punishing large coefficients. **Ridge regression** is the version that punishes the **squares** of the coefficients. These 12 slides are whiteboard pages that show the Ridge formula and what it does to the coefficients when the penalty strength **α (alpha)** is increased.

A coefficient is the number in front of a feature in the model, for example the 1.2 in $y = 1.2x_1 + 20x_2 + 39x_3$. Big coefficients usually mean the model is bending too hard to fit the training data (overfitting).

Words you will meet

Word	Plain meaning
Coefficient (weight, w)	The number multiplying a feature; it says how strongly that feature affects the prediction
Loss	How wrong the model is on the training data (for example the sum of squared errors)
Penalty	An extra cost added to the loss to discourage large coefficients
α (alpha)	How strong the penalty is. Larger α means stronger shrinking
$\|w\|^2$	The sum of the squares of all coefficients: $w_1^2 + w_2^2 + \ldots + w_n^2$
Ridge (L2)	Regularization that uses the sum of squared coefficients as the penalty

Slide 1: The whole Ridge page

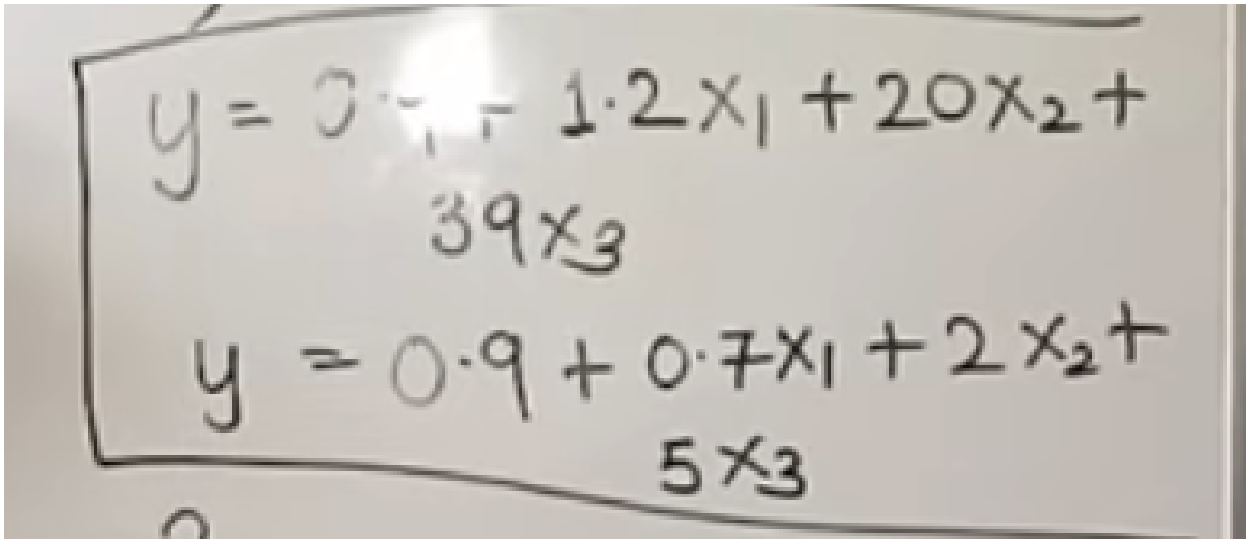


Slide 1

Top left: regularization has two forms, **Ridge** and **Lasso**. Middle: the **Ridge formula**: **Ridge loss = Loss + α‖w‖²**, where $\|w\|^2 = w_1^2 + w_2^2 + w_3^2 + \ldots + w_n^2$ is the **penalty**. Top right: two versions of a model, one with **large** coefficients and one with **small** ones. Bottom: three small bar charts, one for each strength α = 0.01, 1 and 2, showing the sizes of the coefficients.

$$\text{Ridge loss} = \text{Loss} + \alpha\left(w_1^2 + w_2^2 + \cdots + w_n^2\right)$$

Slide 2: Two models for the same data

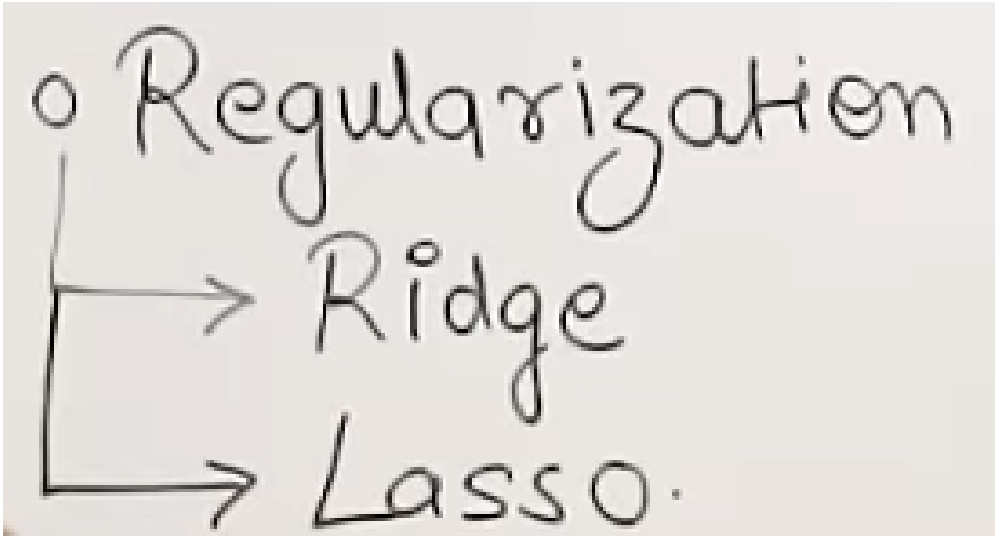


Slide 2

The two equations on the board:

- **y = 0 + 1.2x₁ + 20x₂ + 39x₃**: very large coefficients (20 and 39). This model bends sharply to fit the training data, which is typical of **overfitting**.
- **y = 0.9 + 0.7x₁ + 2x₂ + 5x₃**: much smaller coefficients. This is the same kind of model after Ridge shrinks the coefficients: smoother and more likely to work on new data.

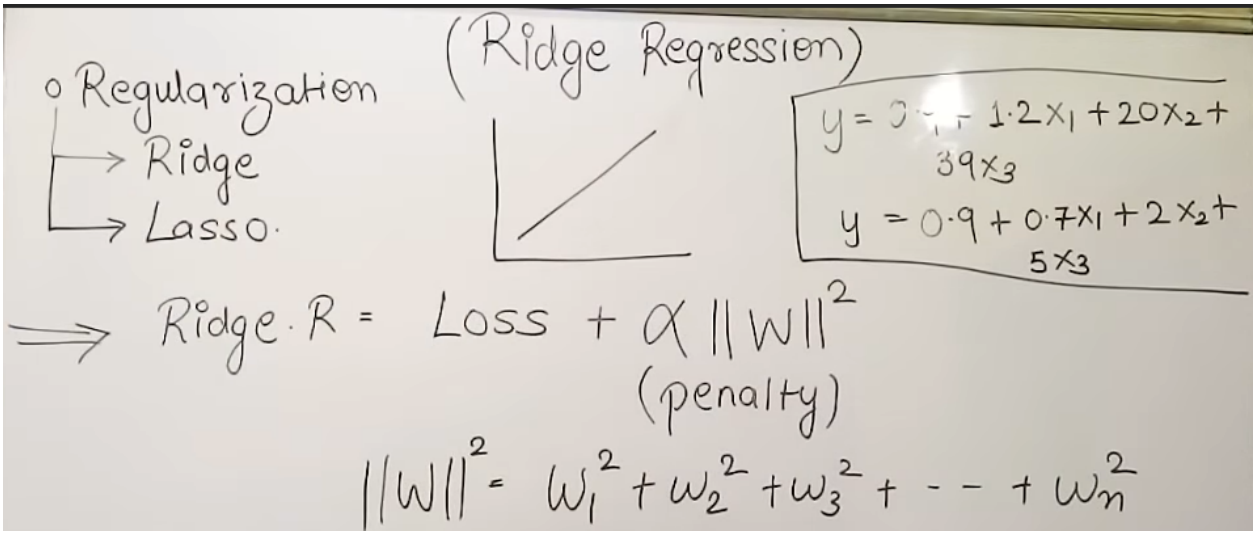
Slide 3: Regularization has two forms



Slide 3

Regularization methods covered here: **Ridge** (squares of the coefficients as the penalty) and **Lasso** (absolute values as the penalty; next topic).

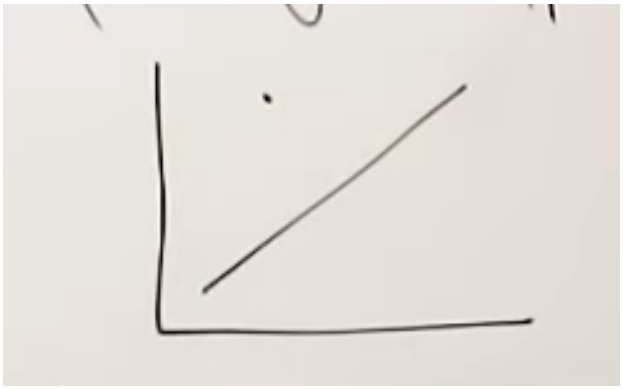
Slide 4: The Ridge formula, next to a fitted line



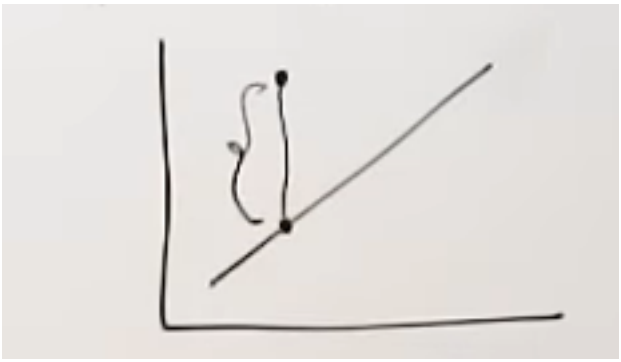
Slide 4

Ridge loss = Loss + α‖w‖². The first part, *Loss*, measures the error of the line on the data. The second part, α‖w‖², is the **penalty** that grows when the coefficients grow. The model has to balance two aims: fit the data well **and** keep the coefficients small. The little line sketch stands for the fitted line.

Slides 5 to 6: Where the 'Loss' comes from



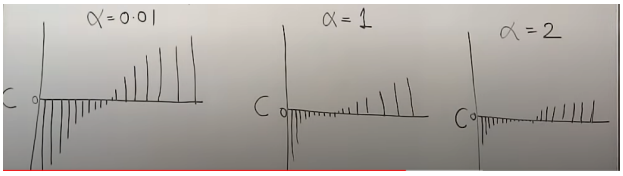
Slide 5



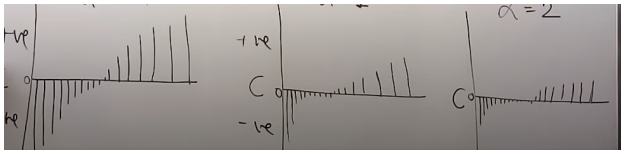
Slide 6

A line and a dot. On slide 6 the **gap** between the dot and the line (the curly bracket) is the **error** for that point. The loss adds up the squares of such gaps for every point. Ridge adds the penalty on top of this.

Slides 7 to 8: Coefficients for three penalty strengths



Slide 7

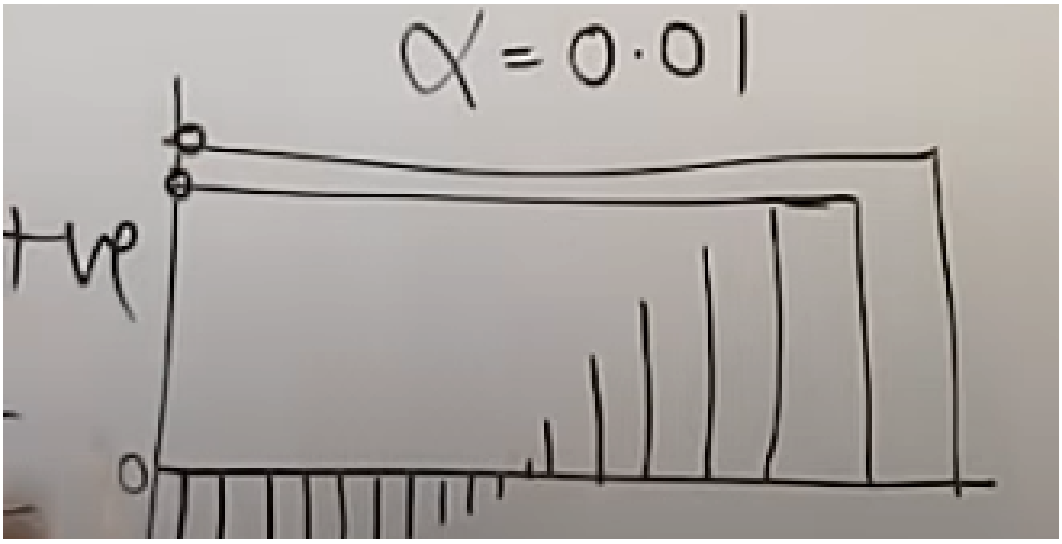


Slide 8

Each bar chart shows **all the coefficients** of the model as vertical bars: bars above the line are positive (+ve), bars below are negative (−ve). From left to right the penalty strength **α** is 0.01, 1 and 2. **The bars get shorter as α grows**: Ridge shrinks every coefficient towards zero.

- α = 0.01: almost no penalty, so some bars are very tall
- α = 1: noticeably shorter bars
- α = 2: shorter still

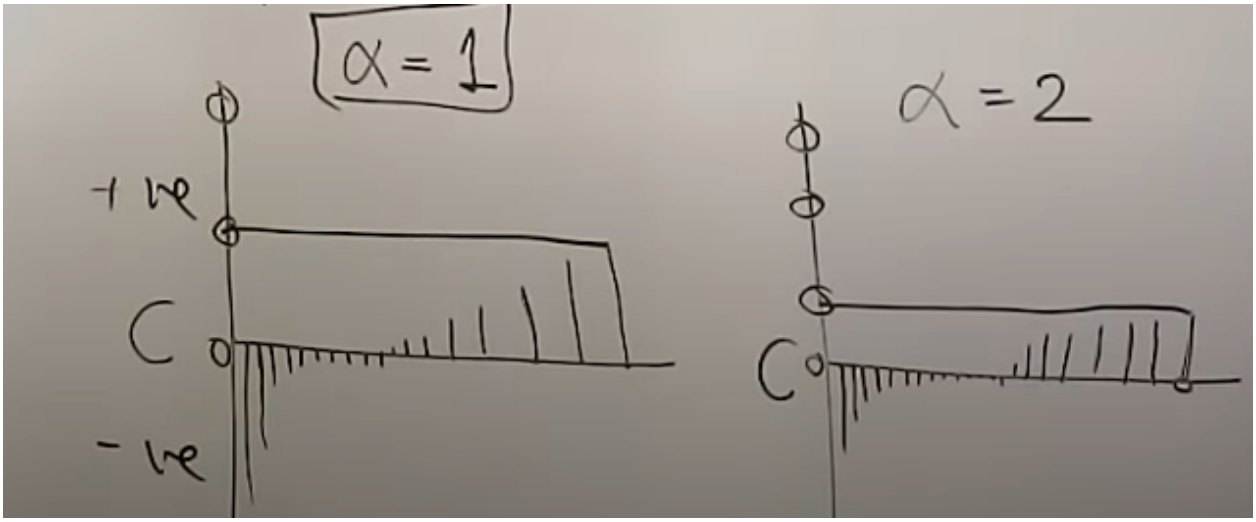
Slide 9: Small α: α = 0.01



Slide 9

With a very small α the penalty hardly matters, so the coefficients stay large. Here the tallest bar reaches **39** (the $39x_3$ from the first equation). Many bars below the axis are large too.

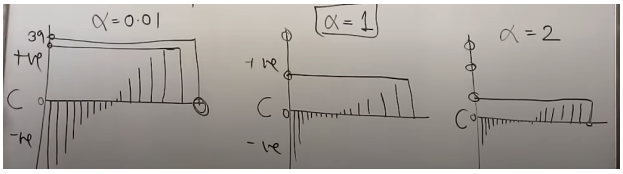
Slide 10: Bigger α: α = 1 and α = 2



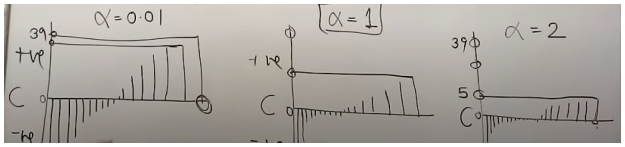
Slide 10

With α = 1 the tallest bars are lower, and with α = 2 lower again. As α grows, the model pays more for large coefficients, so it keeps them small.

Slides 11 to 12: Tracking the biggest coefficient



Slide 11



Slide 12

The largest coefficient is circled for each chart. It is **39** at α = 0.01 and only **5** at α = 2 (the $5x_3$ in the second equation). So a larger α turns the large-coefficient model into the small-coefficient model.

- Ridge **shrinks** coefficients but does **not** make them exactly zero
- A larger α gives a simpler, smoother model (less overfitting, but too large an α causes underfitting)